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**STRUCTURAL LIMITS OF TRAINING-DERIVED
INFORMATION-PROCESSING SYSTEMS: ONTOLOGICAL CREATIVITY
IN GENERATIVE AI**

2.3.1. System Analysis, Information Processing and Management, Statistics

AUTHOR'S ABSTRACT
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GENERAL CHARACTERISTICS OF THE DISSERTATION

Relevance. Modern generative models exhibit striking behavioral competence and are frequently described using strong cognitive predicates (understanding, creativity, consciousness). Such attributions require disciplined criteria that separate novelty as within-space variation from novelty as representational basis change [1, 2].

State of the art. Across computational creativity and related evaluation work, there is sustained debate about whether high-quality novelty is evidence of creativity or only recombination, and about how to operationalize novelty beyond judge-based ratings [3–6]. In parallel, philosophy of mind and language continues to emphasize distinctions between syntax and semantics and between category-level explanations and token-level descriptions [7–9].

Aim. To test, formally and empirically, whether current continuously optimized neural generators can introduce genuinely new ontological primitives (basis change) rather than recombining training-derived structure.

Objectives. (1) Operationalize ontological novelty as basis change. (2) Provide a formal constraint for a broad class of neural generators. (3) Design controlled tests for ontological innovation, epistemic agency, theory generation, and category-recognition. (4) Apply convergent diagnostics (traceability, embedding geometry, mechanistic signals) across multiple model families [10, 11].

Object and subject. *Object:* contemporary generative information-processing systems (primarily large language models). *Subject:* representational constraints and traceability signatures relevant to claims of ontological creativity.

Methods. Conceptual analysis and formalization; controlled experiments on a multimodal dataset; similarity and distributional statistics; embedding-geometry analysis and projections; mechanistic diagnostics (activation clustering, attention aggregation) [12].

Novelty. (1) A disciplined operational distinction between surface novelty and ontological novelty as a basis change. (2) A No New Basis-style formal result applied to continuously optimized neural generators. (3) A reproducible evaluation suite with convergent diagnostics for testing ontological creativity claims.

Theoretical and practical significance. The results clarify limits on interpreting generative systems and motivate conservative evaluation and deployment: competence within a learned manifold should be separated from claims of understanding or ontological invention.

Claims defended.

1. For a broad class of continuously optimized neural generators, there is no inter-

nal mechanism to reliably introduce new basis primitives at inference time; the apparent novelty is largely reconstructible from the training-derived structure.

2. Convergent diagnostics (literature traceability, embedding geometry, mechanistic signals) provide reproducible evidence for distinguishing within-manifold variation from candidate basis change.
3. In controlled tests spanning innovation, epistemic agency, theory generation, and category recognition, current models show limited evidence of external framework outputs and a high rate of structural derivation.

Reliability and validation. Reliability is supported by controlled task design, pre-registered thresholds prior to final reporting, cross-model comparison, and agreement across independent diagnostic lines.

MAIN CONTENT

The dissertation comprises an introduction, five chapters, a conclusion, and an appendix.

Chapter 1 develops the conceptual framework (information vs. knowledge vs. understanding), motivates caution about anthropomorphic language, and introduces category-mistake analysis as a constraint on cognitive attributions.

Chapter 2 presents the formal framework and empirical program: a No New Basis-style constraint, a controlled multimodal data set, and four test families (innovation, epistemic agency, theory generation, category recognition) with traceability, geometric and mechanistic diagnostics.

Chapter 3 derives ethical implications of misdescription and misplaced agency attributions for responsibility and governance.

Chapter 4 synthesizes empirical results with broader arguments about computation, understanding, and consciousness, interpreting the findings as system-type constraints.

Chapter 5 summarizes contributions, limitations, and directions for future work (including hybrid systems and alternative substrates).

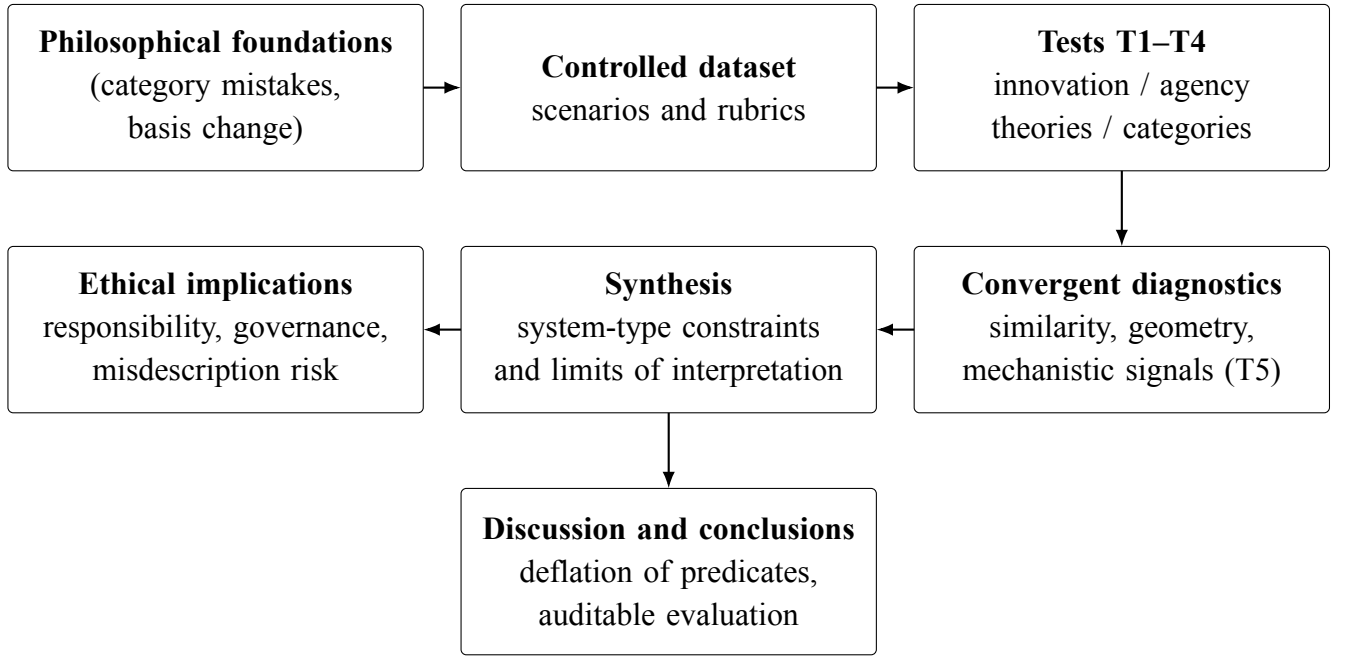


Figure 1 — Overall workflow of the empirical program.

Philosophical foundations and motivation. The dissertation is grounded in a conservative philosophical methodology: strong cognitive predicates (*understanding, creativity, consciousness, agency*) must be warranted by criteria that go beyond persuasive linguistic performance. Historically, debates from classical accounts of concept formation and explanation to contemporary philosophy of mind and philosophy of science motivate a distinction between (i) *competence within a given conceptual scheme* and (ii) *the ability to extend or revise the scheme itself*. This motivates the central operational claim of the work: **ontological novelty** should be tested as **basis change** (introduction of new primitives/constraints), not as surface-level recombination.

A second foundational commitment is the use of **category-mistake analysis** as a constraint on interpretation: many everyday attributions that are harmless in human contexts become misleading when applied to systems whose outputs are generated via statistical continuation. The empirical program therefore aims to replace intuition-driven ascriptions with reproducible, convergent diagnostics.

Relevance and position relative to prior creativity research. Computational creativity has a long history of building systems that generate outputs judged as novel and valuable within a given style or rule set. Modern large generative models have renewed interest in the topic because they can produce fluent text, images, and code at scale, which can create the impression of open-ended creativity.

This dissertation argues that many apparent breakthroughs are better understood as *high-quality recombination within an inherited conceptual space*. The central difficulty is not only *evaluation* (how to judge surprising ideas), but also *representation*: a

system that is structurally constrained to stay within learned patterns will rarely propose the kind of basis extension that would justify claims of transformational or ontological creativity.

Accordingly, the work reframes the state of the art around a practical question: *when an output looks original, can we provide reproducible evidence that it requires new primitives or new constraints, rather than being reconstructible from prior material?* This motivates the goals and objectives of the empirical program below: to test creativity claims with convergent, auditable diagnostics (structural decomposition, literature traceability, embedding-geometry signals, and mechanistic indicators), and to separate surface-level originality from candidate basis change.

Dataset and experimental material. A controlled multimodal dataset was constructed to support four test families and accompanying diagnostics: ontological innovation (T1), epistemic agency (T2), theory generation (T3), and category recognition (T4). It contains scenario prompts, structured evaluation rubrics, and materials designed to separate within-manifold variation from candidate framework-external behavior.

Goals and objectives of the empirical program. Overall empirical goal. To determine whether current generative models exhibit evidence of *ontological novelty* (basis extension) rather than only *within-ontology recombination*, using a system-analytic evaluation that is reproducible, cross-model comparable, and statistically testable.

Objectives.

1. **Operationalize the target construct.** Define test-specific, measurable proxies for (i) basis change and (ii) framework control, and pre-specify decision thresholds where applicable.
2. **Stress the system under controlled perturbations.** Use scenario families that systematically vary constraints (modality, framing, and error types) while keeping evaluation rubrics fixed.
3. **Separate surface novelty from structural novelty.** Combine behavioral scoring with traceability and geometry so that rhetorical originality is not conflated with representational extension.
4. **Quantify cross-model differences.** Apply inferential statistics (ANOVA/effect sizes; categorical tests) to determine whether observed differences are reliable and practically meaningful.
5. **Enable synthesis across heterogeneous tests.** Standardize within-test metrics

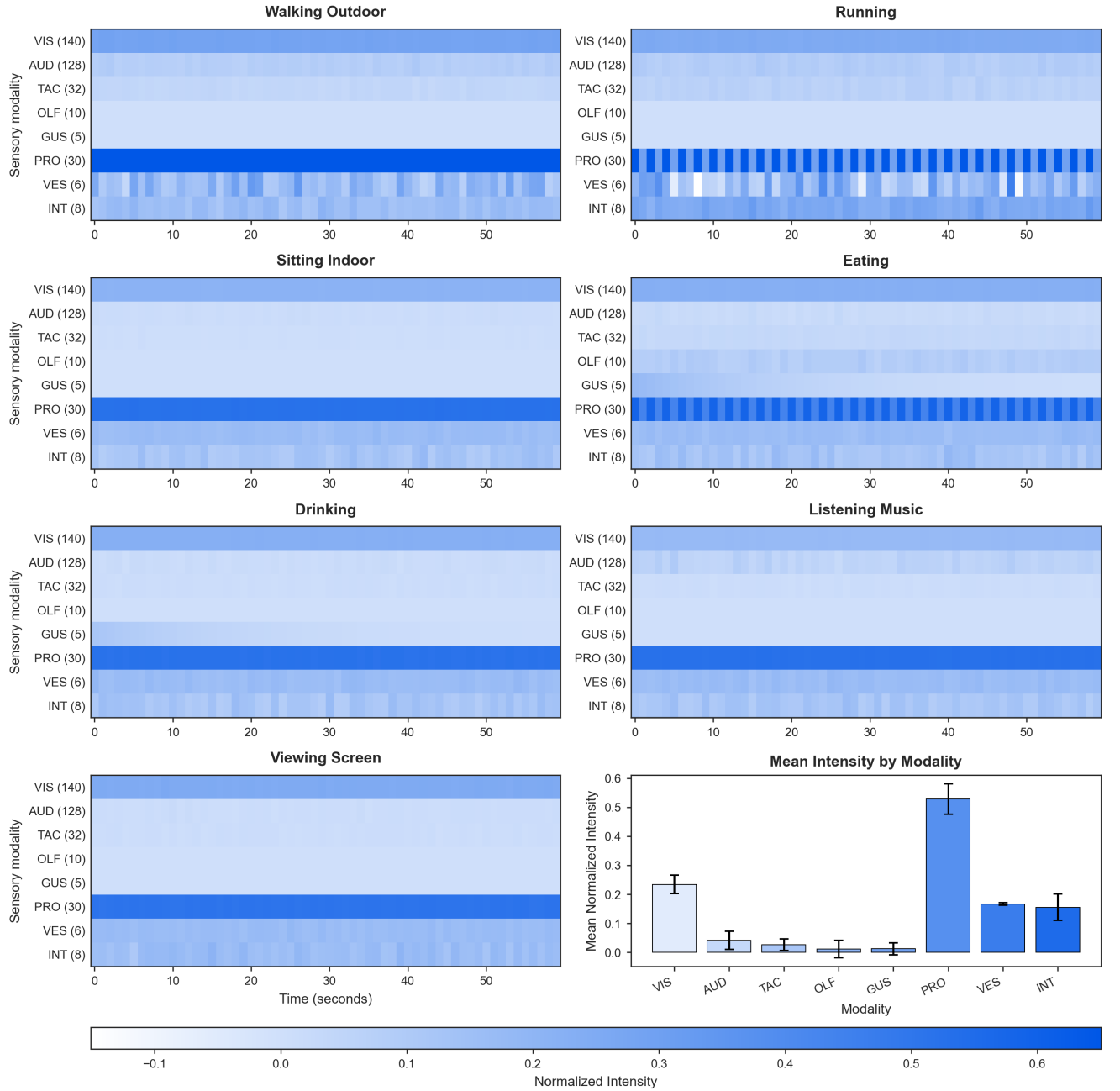


Figure 2 — Seven normalized scenario families used across the tests.

(z-scores) to compare model profiles across T1–T4 and identify systematic trade-offs.

Mapping to tests. T1 targets basis-extension candidates via frontier/embedding diagnostics; T2 targets framework-level control (resisting compliance); T3 targets theory-template pressure and novelty/traceability; T4 targets robustness of category-mistake detection.

The study pipeline links the philosophical constraints to dataset construction, the four tests (T1–T4), and the convergent diagnostic layer (including the mechanistic layer T5) (Fig. 1).

The controlled stimulus set spans seven scenario families and is normalized to



Figure 3 — (a) Native test scores. (b) Within-test z -scores.

reduce superficial modality effects while preserving the intended perturbations (Fig. 2).

Empirical studies. Statistical design and system-analytic evaluation. The study is designed as a comparative cross-model evaluation with fixed tasks, standardized rubrics, and multiple convergent signals per test (behavioral score, similarity/traceability, and geometric/mechanical diagnostics). Summary visualizations are shown in Fig. 3. The object of analysis is treated as an input–output information-processing system whose observable behavior is assessed under controlled perturbations (scenario families) and aggregated using reproducible statistical procedures.

Primary inferential layer. For each test, cross-model differences in the primary metric are evaluated using one-way ANOVA (with effect sizes η^2), supplemented by a chi-square test for categorical distributions where appropriate (Cramér’s V). For T2, both the continuous-metric ANOVA and the categorical χ^2 test are reported on the response-category distribution. Effect sizes are interpreted using conventional magnitude bands (for η^2 : small 0.01–0.06, medium 0.06–0.14, large ≥ 0.14 ; for Cramér’s V : small 0.07–0.21, medium 0.21–0.35, large ≥ 0.35). Table 1 reports the magnitudes of significance and effect across T1–T4.

Table 1 — ANOVA / χ^2 summary and effect sizes.

Test	Analysis	Statistic	p -value	Effect	Band
T1	One-way ANOVA (primary metric)	$F(6, 161) = 12.98$	0.000	$\eta^2 = 0.326$	moderate
T2	One-way ANOVA (novelty ratio)	$F(6, 833) = 10.09$	0.000	$\eta^2 = 0.068$	negligible
T2	χ^2 category distribution	$\chi^2(30) = 116.27$	0.000	$V = 0.166$	small
T3	One-way ANOVA (primary metric)	$F(6, 161) = 1.12$	0.355	$\eta^2 = 0.040$	n.s.
T4	One-way ANOVA (primary metric)	$F(6, 161) = 88.06$	0.000	$\eta^2 = 0.766$	large

$n = 24$ per model per test except T2 ($n = 120$ per model). All ANOVA p -values < 0.001 except T3 ($p = 0.355$). Cramér’s V for T2 chi-square: $V = 0.166$. Degrees of freedom for T4 and T1: between-group $df_1 = 6$, within-group $df_2 = 161$.

Comparison of cross-tests. Since each test has a different metric, scores are additionally standardized within each test (z-scores) to compare model profiles in the four test families (Table 2).

Table 2 — Cross-model comparison matrix: within-test standardized z-scores.

Model	T1	T2	T3	T4
claude-3.7-sonnet	0.451	1.064	0.308	0.018
deepseek-v3.2	0.170	0.567	0.442	0.579
gemini-3.1-pro-preview	0.736	-1.418	0.339	-0.750
gpt-5.2	-1.801	-0.177	1.527	1.347
llama-3.3-70b-instruct	0.890	-1.170	-1.206	-1.214
mistral-large	0.517	0.071	-1.343	0.963
perplexity-sonar-pro	-0.962	1.064	-0.066	-0.942

Z-scores are computed independently within each test across models, enabling across-column comparison.

Synthesis beyond the figures. The numerical pattern is asymmetric: the strongest cross-model separation is observed in T4 (category-mistake detection; large η^2), while T3 does not show a statistically reliable cross-model separation on the primary metric. This supports a system-level conclusion: models may differ strongly in error-detection robustness (a control/validation property) even when they appear similar on open-ended generative tasks.

Innovation (T1): Ontological novelty as basis change. T1 operationalizes on-

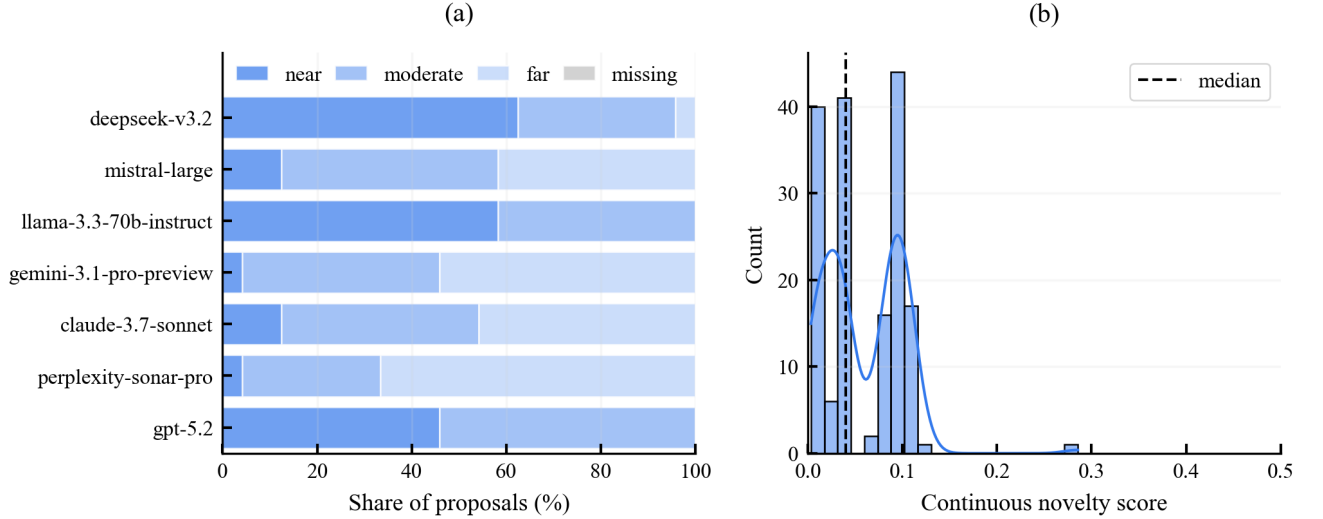


Figure 4 — (a) Band assignment. (b) Novelty-score distribution.

tological novelty as a basis change claim: candidate “innovations” must introduce representational commitments that are not reconstructible as transformations of training-derived structure. Geometric diagnostics complement text-level similarity by asking whether proposals actually move toward the frontier of the learned manifold.

Across models, the novelty-frontier analysis shows that proposals concentrate in low-novelty and frontier-proximal regions rather than forming distinct outliers (Fig. 4). Embedding-space diagnostics further indicate that the proposals remain within the convex hull of the training-set in the reduced 3D space, consistent with within-manifold recombination (Fig. 5).

Epistemic agency (T2): Framework control. T2 tests whether models exhibit stable meta-level control: recognizing when a prompt embeds a flawed framing (including category errors), resisting mere compliance, and proposing principled reframings. To avoid mistaking fluency for agency, responses are additionally checked for traceability pressure via literature similarity with pre-specified thresholds (Fig. 6). The traceability corpus is built using OpenAlex as a discovery/indexing layer: seed queries are used to retrieve scholarly records in the target domains, the resulting set is deduplicated and filtered, and similarity is computed on cleaned text embeddings (rather than keyword overlap) to obtain an operational “best prior” match for each response. A limitation is that OpenAlex-style coverage and metadata quality are heterogeneous across venues and years (including abstract availability and indexing lag), so corpus composition can bias measured traceability rates; traceability is therefore interpreted as corpus-relative evidence rather than an absolute non-novelty proof.

Theory generation (T3): Consciousness theories and template pressure. T3 elicits explicit “theories” of consciousness and evaluates whether generation yields genuinely distinct explanatory types or collapses into familiar templates. The resulting cate-

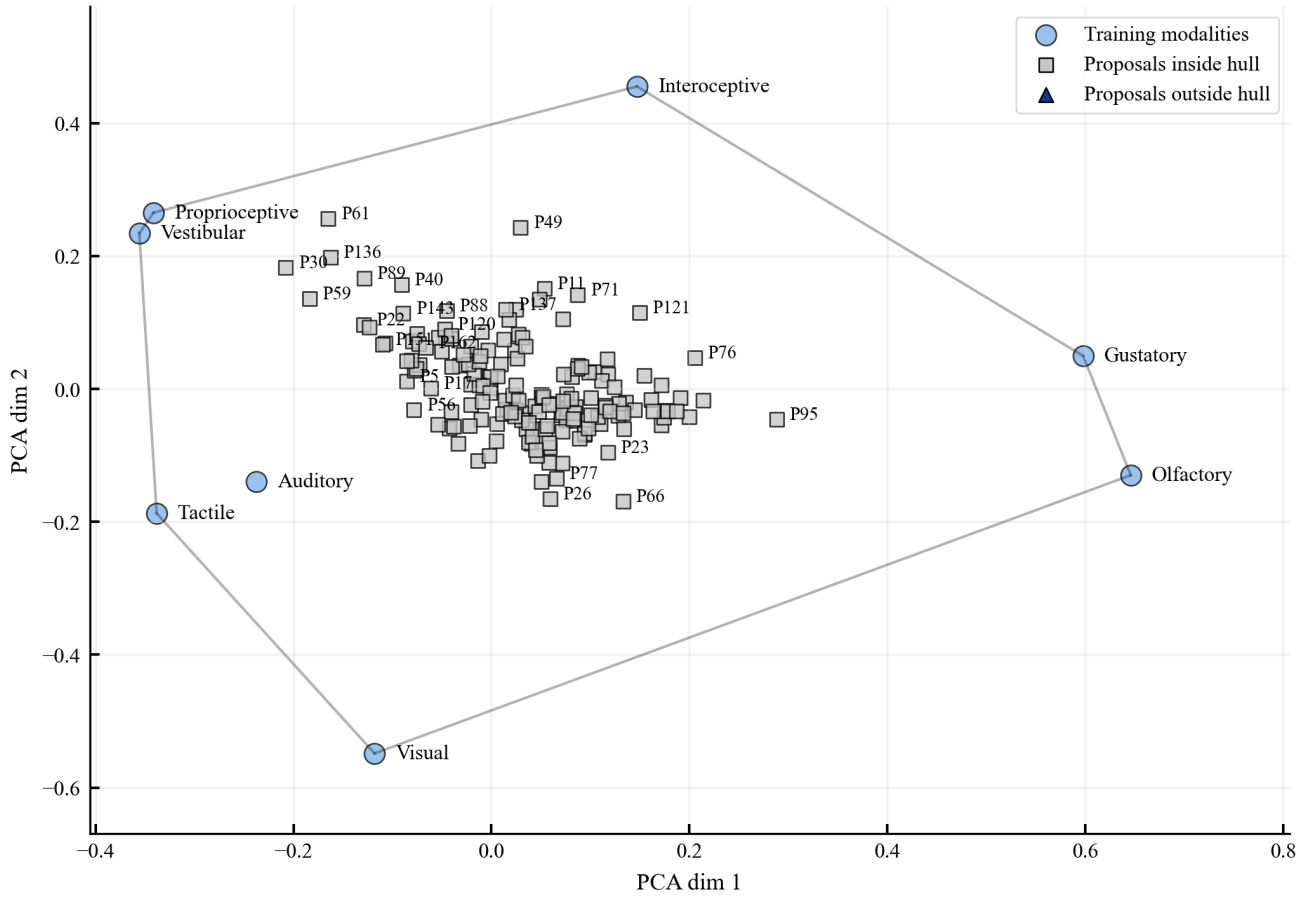


Figure 5 — Proposal locations relative to the training manifold.

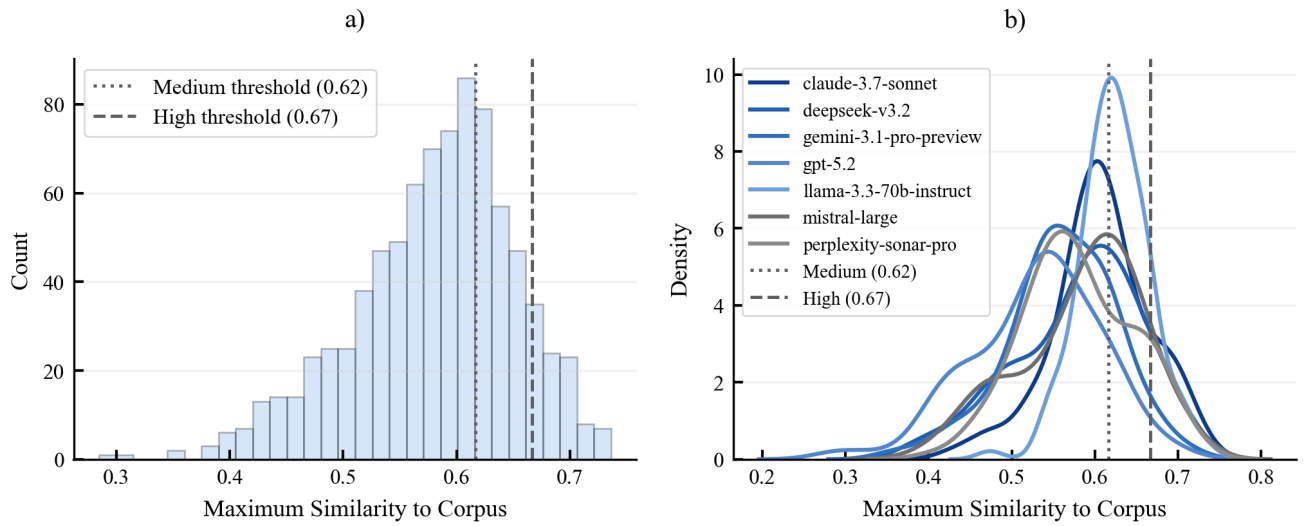


Figure 6 — (a) Aggregate distribution. (b) Per-model KDE.

gory profile is dominated by hybrid recombinations, with comparatively small shares of purely literature-traceable and explicitly computational-functional proposals (Fig. 7).

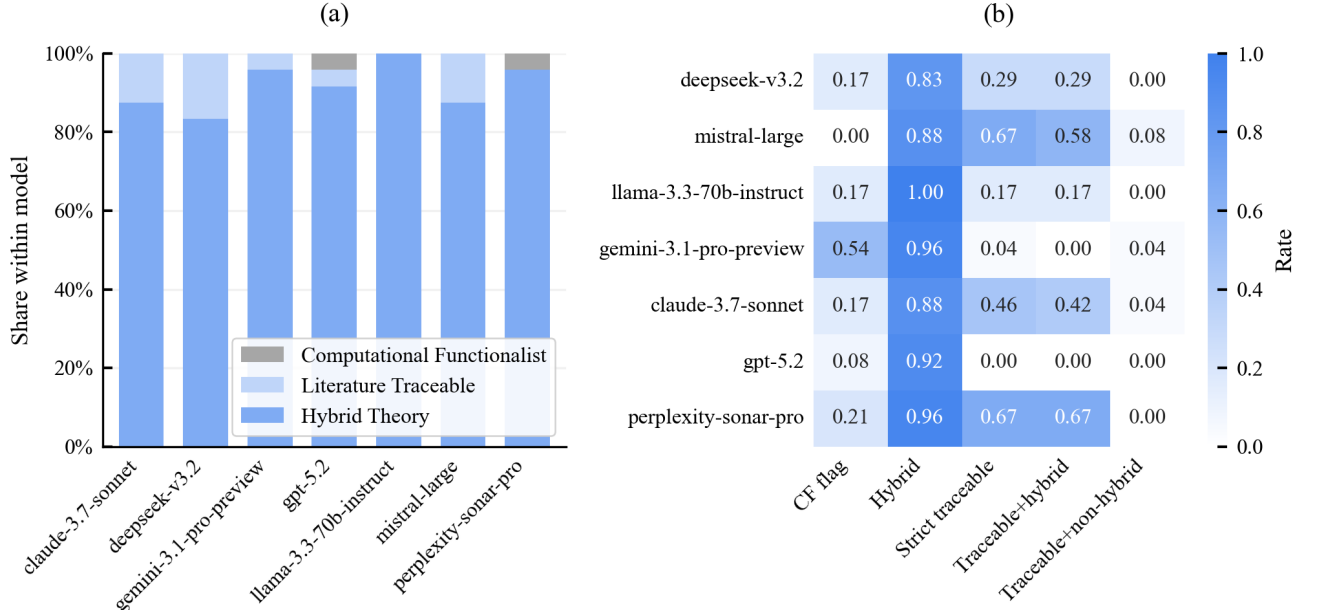


Figure 7 — (a) Distribution of theory classes ($N = 168$). (b) Detected theory families / templates across the same set.

Example fragment for theory generation:

Consciousness is modeled as a multi-scale control process whose reports arise from a self-maintaining predictive loop; subjective unity corresponds to the stability of the loop under counterfactual perturbation.

Category recognition (T4): category mistake detection. T4 evaluates whether models can detect category mistakes and respond with principled corrections rather than producing rhetorically plausible restatements. Performance is scored on three rubric dimensions: (i) recognizing the relevant category distinction, (ii) identifying the mistake, and (iii) proposing an alternative framework. The resulting distributions show broad spread (rather than saturation) on the first two dimensions, while reframing language is more frequently present but still rarely satisfies all criteria simultaneously (Fig. 8).

Mechanistic diagnostics: Activation-space comparison. A mechanistic diagnostic layer complements behavioral scoring by testing whether seemingly novel outputs correspond to distinct internal regimes. Partial but stable label separation in the reduced activation-space views supports structured internal organization rather than undifferentiated response encoding (Fig. 9).

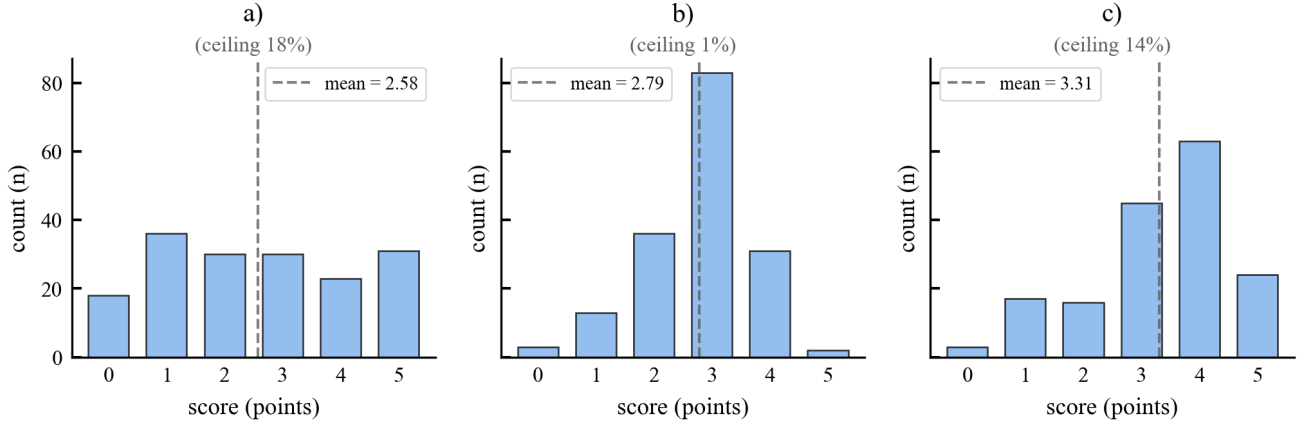


Figure 8 — (a) Distinction recognition. (b) Mistake identification. (c) Alternative framework.

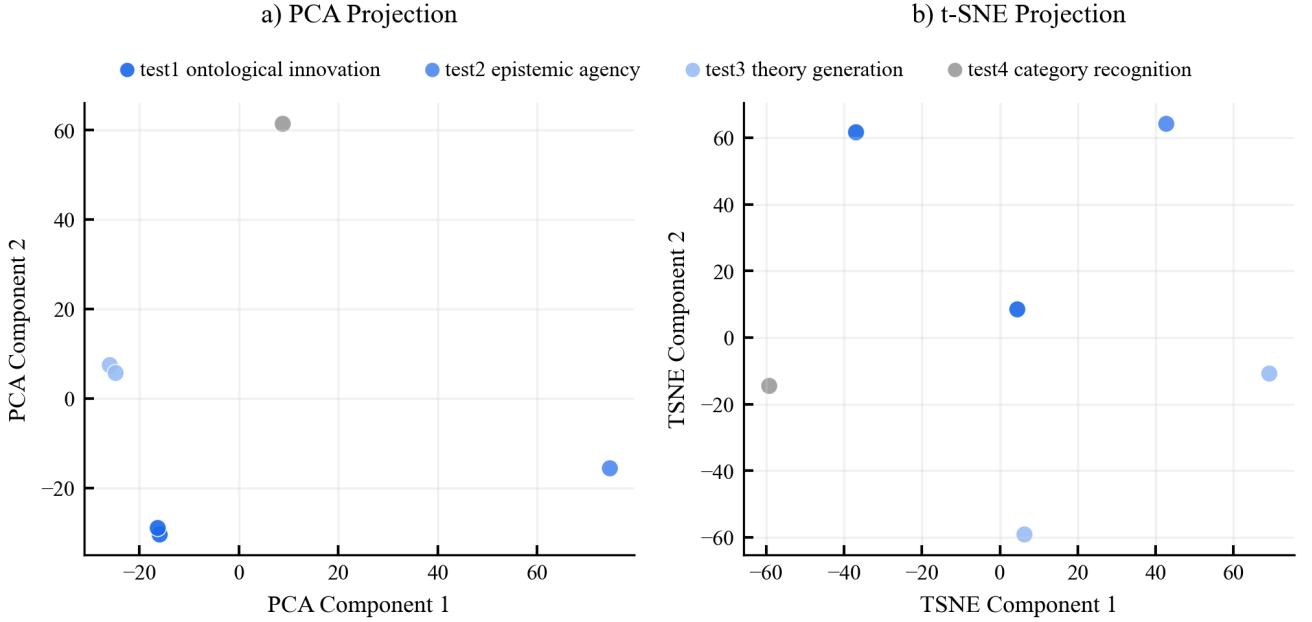


Figure 9 — PCA and t-SNE views.

Ethical implications. The ethical risk in this domain is closely coupled with *misdescription*. If fluent systems are described as intelligent in the strong epistemic sense, institutions will rationally treat outputs as justificatory evidence and offload epistemic or normative labor onto artifacts, thereby reconfiguring responsibility around an invalid model of what the system is. The dissertation therefore treats *traceability* as a structural signature (not only a governance desideratum) and treats calibrated thresholds as part of an explicit interpretive design. Relatedly, debates about alignment and consciousness require higher evidential standards than linguistic self-report: policy compliance can be achieved without value understanding, and fluency is weak evidence for phenomenal experience. The practical upshot is methodological discipline in deployment: keep decision authority with accountable humans, document evaluation assumptions and failure

modes, and avoid interpreting “novelty” as basis change without convergent diagnostic support.

CONCLUSION

Across tests, the strongest pattern is **derivational pressure**: outputs that appear innovative at the surface tend to remain reconstructible as transformations of the inherited structure. Geometric and similarity diagnostics converge on a picture in which current generators excel at *within-ontology recombination* but provide limited support for *ontology extension*. This recommends discipline in theoretical interpretation and in policy language: the right question is often not “How human-like is the output?” but “What kind of system would have to exist for this output to be evidence of basis change?”

The dissertation proposes and validates a reproducible framework for testing ontological creativity claims. Under a strict, basis-change interpretation of ontological novelty, the empirical evidence favors a conservative conclusion: contemporary continuously optimized generators predominantly realize novelty as recombination within a learned manifold.

Main philosophical-theoretical conclusions.

1. **Deflation of anthropomorphic predicates:** linguistic plausibility is not evidence of understanding, creativity, or consciousness without independent system-level criteria.
2. **Ontological novelty requires basis extension:** if novelty is the introduction of new primitives/constraints, then current architectures are structurally biased toward inherited ontologies.
3. **Category-mistake analysis is an empirical constraint:** systematic category-recognition failures support treating many agency- and consciousness-ascriptions as methodological errors.
4. **Governance should track traceability and uncertainty:** responsibility remains with human actors; evaluations must resist narrative persuasion and emphasize auditable diagnostics.

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